Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (Benzene)
Basis set   : external-CI
Charge   : 0
Multiplicity : 1
Total e-   : 6
n-body order : 4

GROUND STATE ENERGIES

E0 (Ha) : nan
E1 (Ha) : nan
Gap (Ha) : nan
HF (Ha) : nan

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 20

C_LB : coherence witness bound
C_gcg: Conjugate Gradient(Alg. A)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 0
Frozen e-   : 0
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.058749
<D> / site : 0.176458

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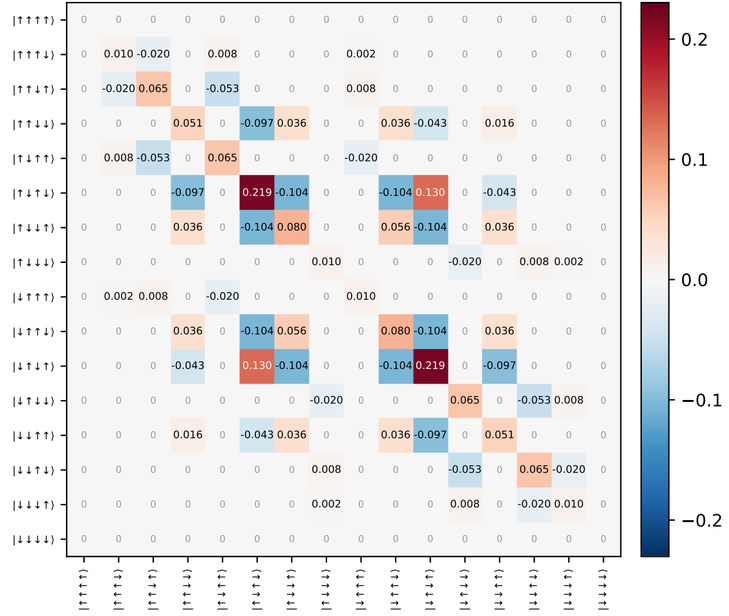
Body order : 4
Clusters   : 15

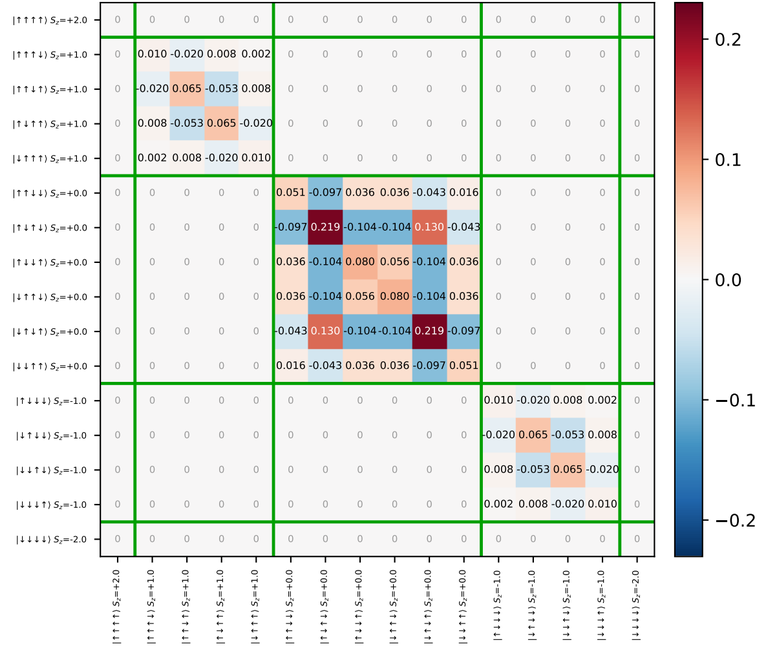
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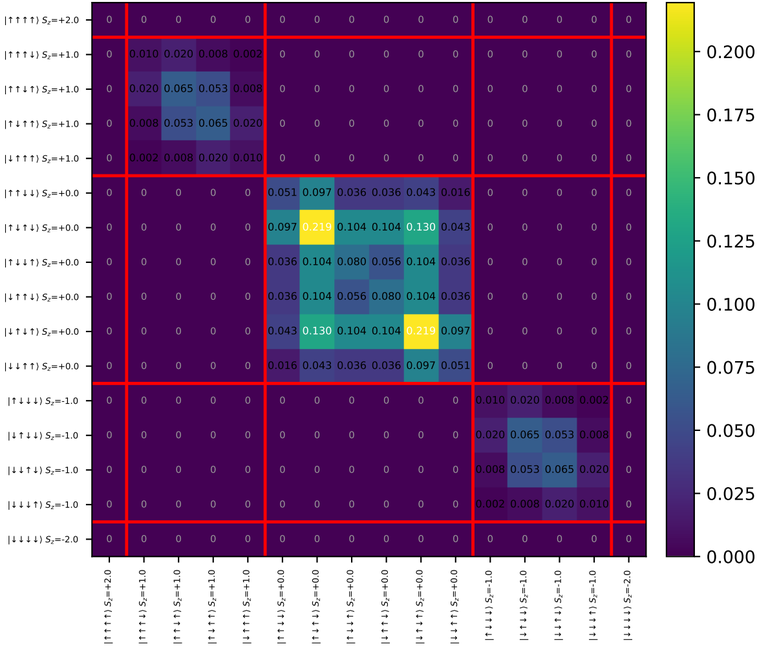
Summary — 4-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2,3	0.329644	1.492429	—	0.000000	0.058735	0.603289	MIXED
0,1,2,4	0.178426	1.920851	—	0.000000	0.015909	0.432503	STRONGLY MIXED
0,1,2,5	0.329636	1.492443	—	0.000000	0.058575	0.603298	MIXED
0,1,3,4	0.166656	2.049886	—	0.000000	0.023082	0.740865	STRONGLY MIXED
0,1,3,5	0.178422	1.920872	—	0.000000	0.019124	0.432500	STRONGLY MIXED
0,1,4,5	0.329644	1.492439	—	0.000000	0.058533	0.603297	MIXED
0,2,3,4	0.178424	1.920864	—	0.000000	0.017324	0.432505	STRONGLY MIXED
0,2,3,5	0.166652	2.049904	—	0.000000	0.026478	0.740861	STRONGLY MIXED
0,2,4,5	0.178422	1.920872	—	0.000000	0.013390	0.432500	STRONGLY MIXED
0,3,4,5	0.329636	1.492443	—	0.000000	0.058692	0.603298	MIXED
1,2,3,4	0.329646	1.492422	—	0.000000	0.058444	0.603294	MIXED
1,2,3,5	0.178424	1.920864	—	0.000000	0.007188	0.432505	STRONGLY MIXED
1,2,4,5	0.166656	2.049886	—	0.000000	0.020211	0.740865	STRONGLY MIXED
1,3,4,5	0.178426	1.920851	—	0.000000	0.013201	0.432503	STRONGLY MIXED
2,3,4,5	0.329644	1.492429	—	0.000000	0.058722	0.603289	MIXED

4-Body RDM (Sites 0,1,2,3) — same matrix in all three views



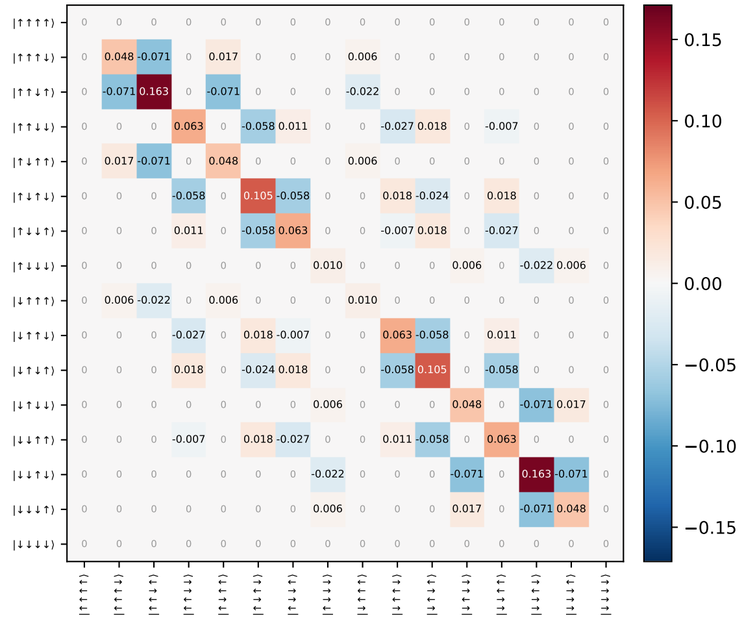


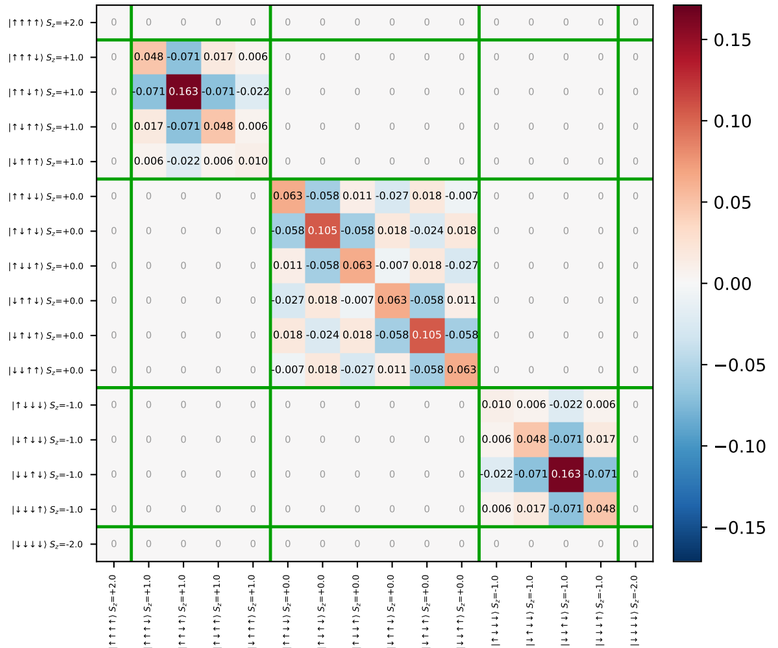


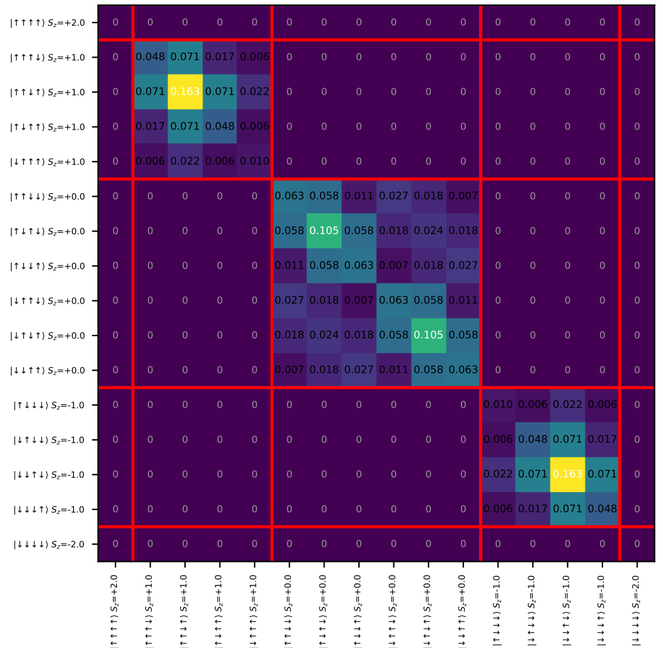
Trace=1.000000 Purity=0.329644 Entropy=1.492429 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.058735 ≤ C_{UB} = 0.603289 verdict: GME

4-Body RDM (Sites 0,1,2,4) — same matrix in all three views

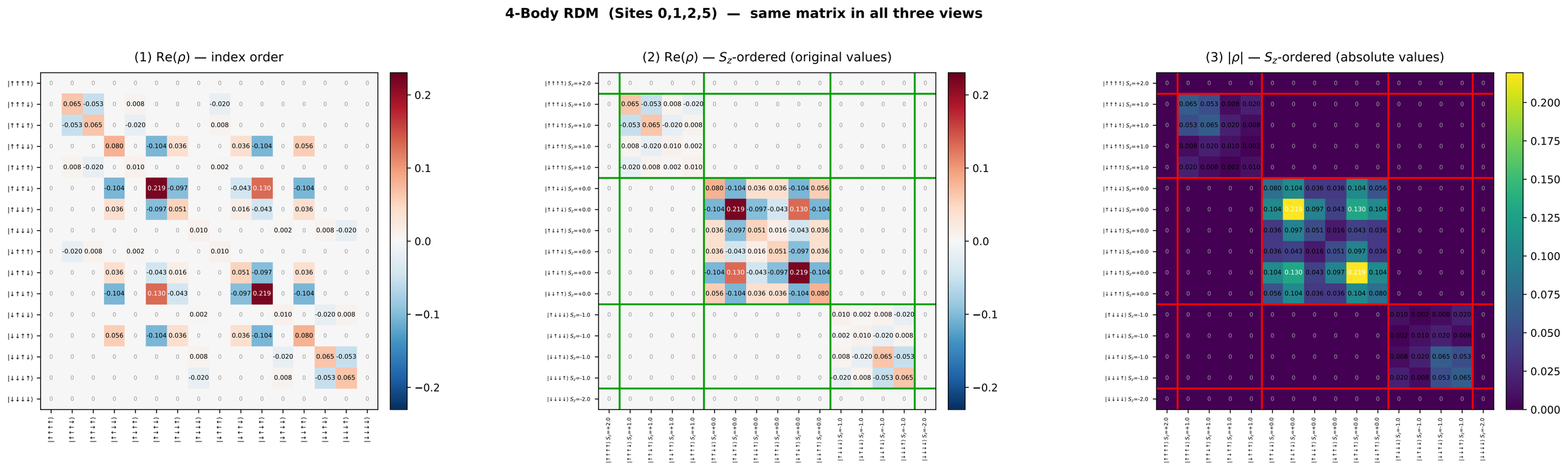






Trace=1.000000 Purity=0.178426 Entropy=1.920851 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

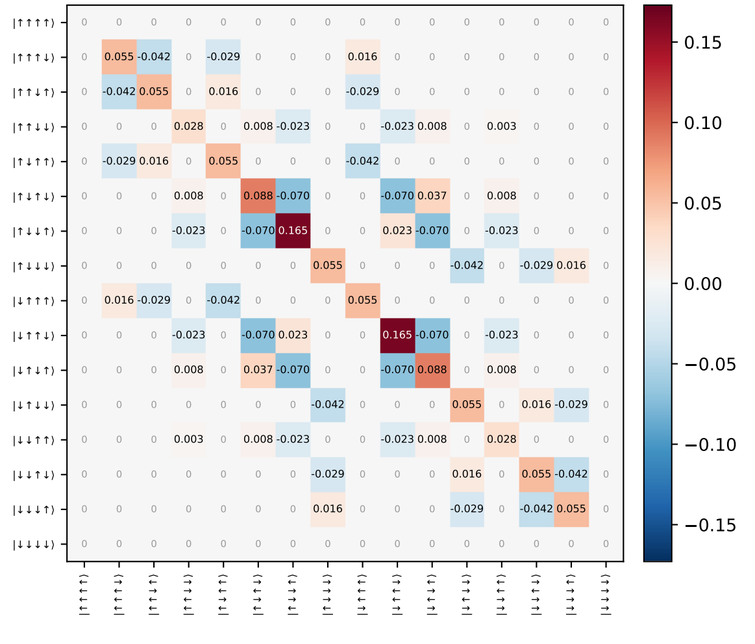
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.015909 ≤ C_{UB} = 0.432503 verdict: GME

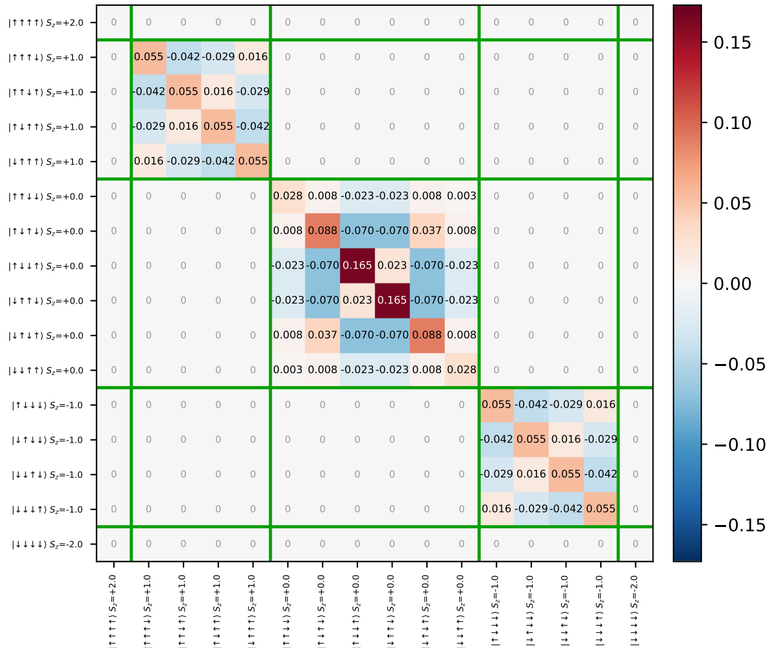


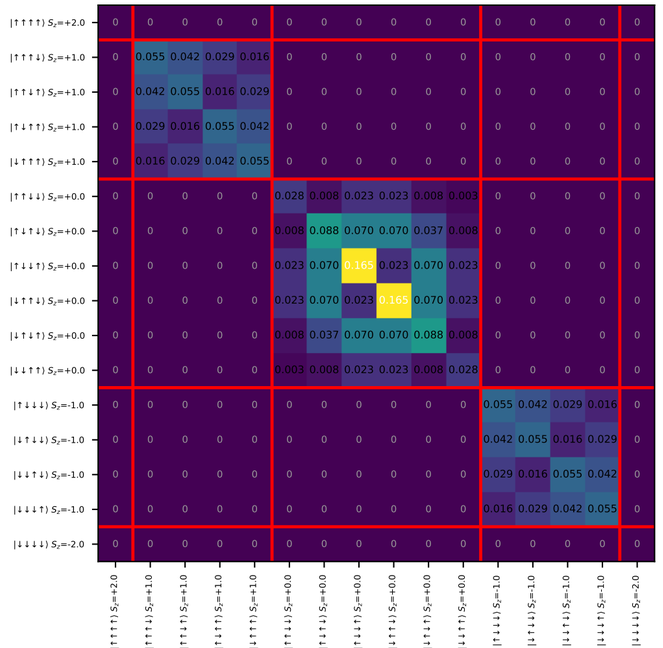
Trace=1.000000 Purity=0.329636 Entropy=1.492443 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.058575 ≤ C_{UB} = 0.603298 verdict: GME

4-Body RDM (Sites 0,1,3,4) — same matrix in all three views



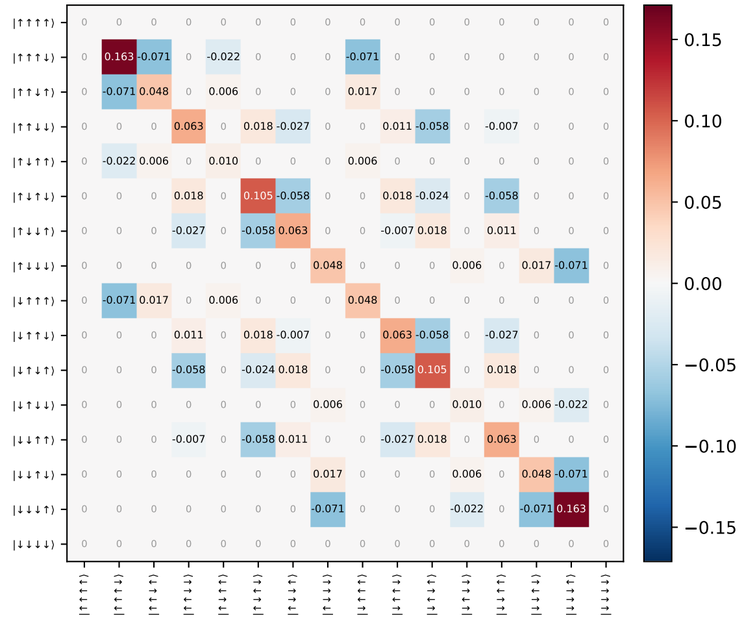


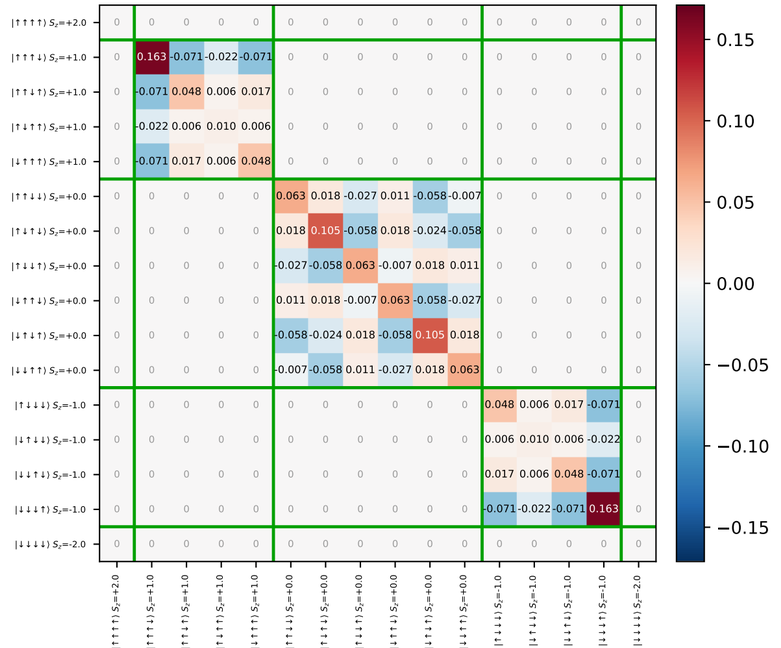


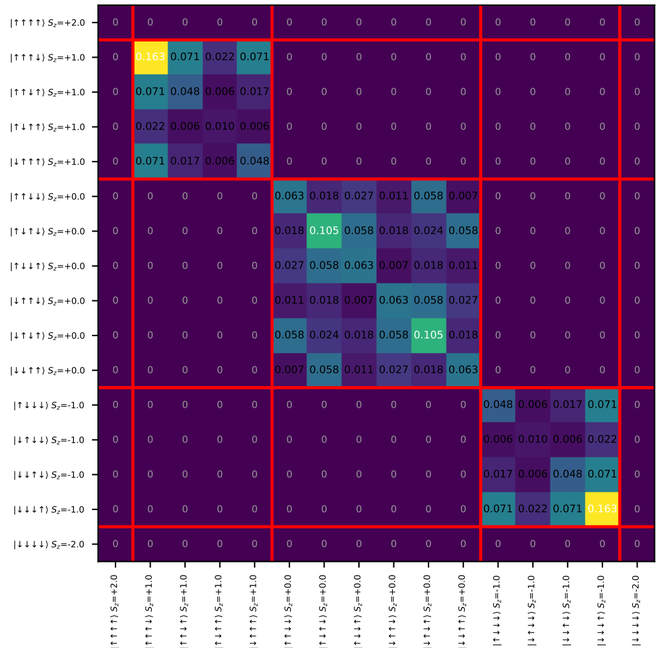
Trace=1.000000 Purity=0.166656 Entropy=2.049886 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.023082 ≤ C_{UB} = 0.740865 verdict: GME

4-Body RDM (Sites 0,1,3,5) — same matrix in all three views





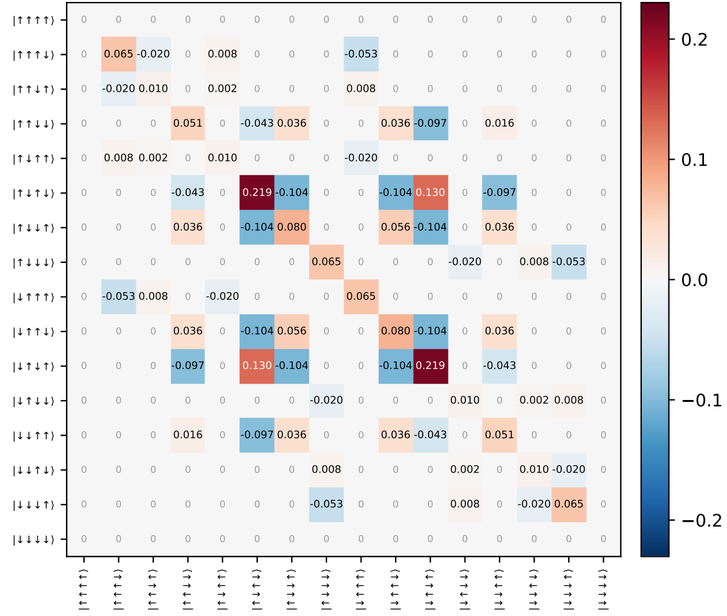


Trace=1.000000 Purity=0.178422 Entropy=1.920872 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

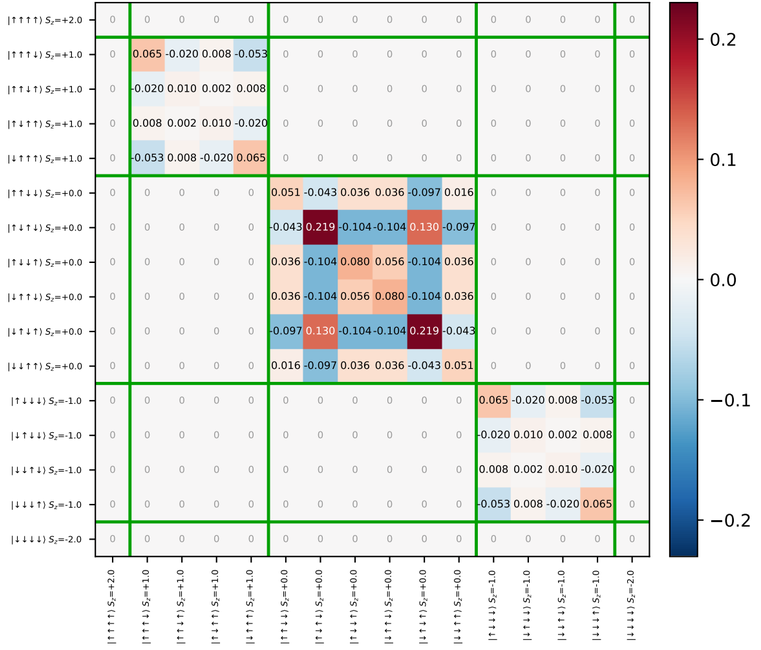
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.019124 ≤ C_{UB} = 0.432500 verdict: GME

4-Body RDM (Sites 0,1,4,5) — same matrix in all three views

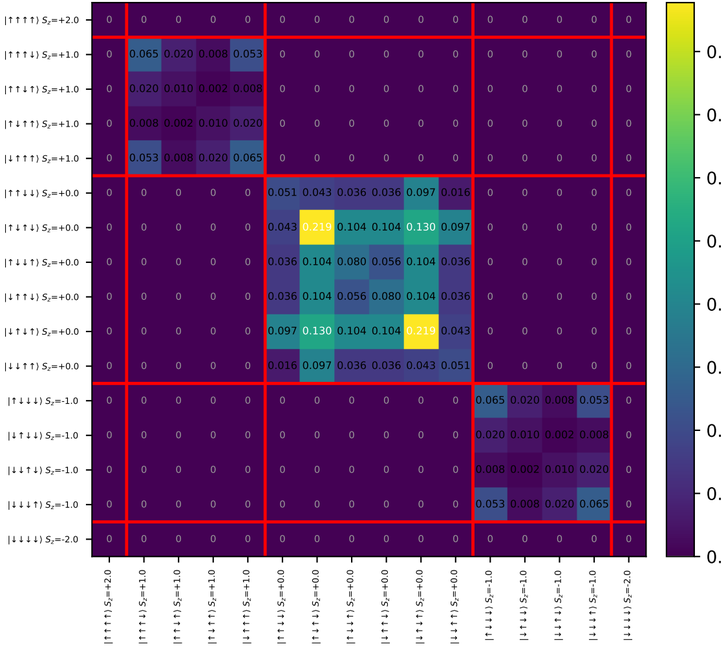
(1) Re(ρ) — index order



(2) Re(ρ) — S_z-ordered (original values)



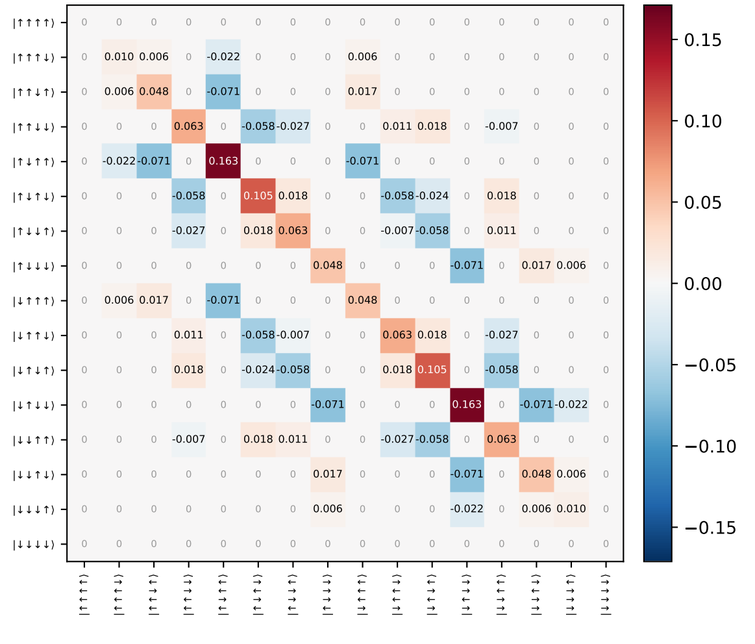
(3) $|\rho|$ — S_z-ordered (absolute values)

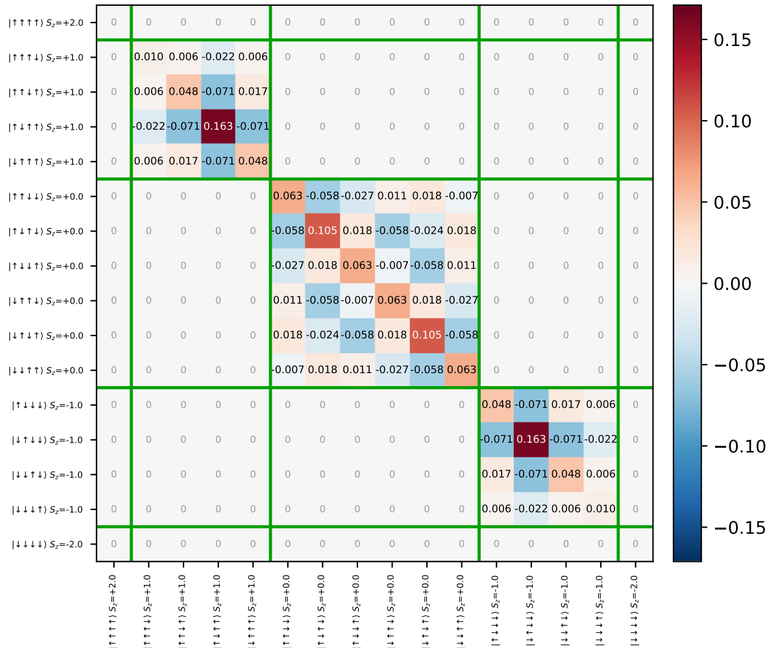


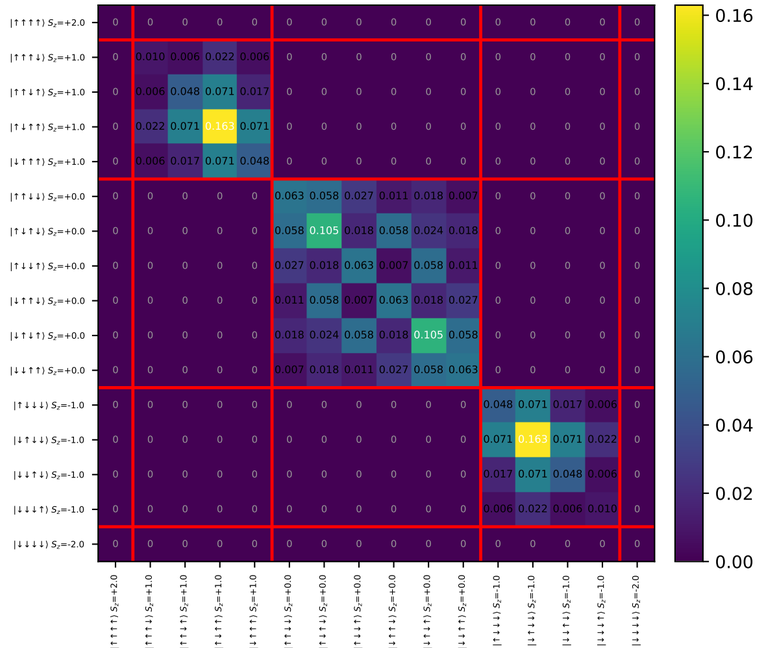
Trace=1.000000 Purity=0.329644 Entropy=1.492439 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.058533 ≤ C_{UB} = 0.603297 verdict: GME

4-Body RDM (Sites 0,2,3,4) — same matrix in all three views



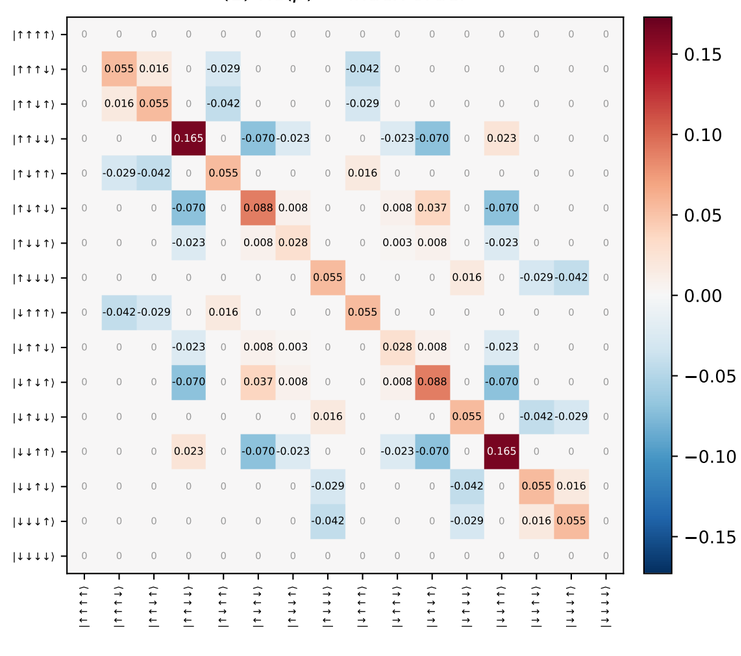


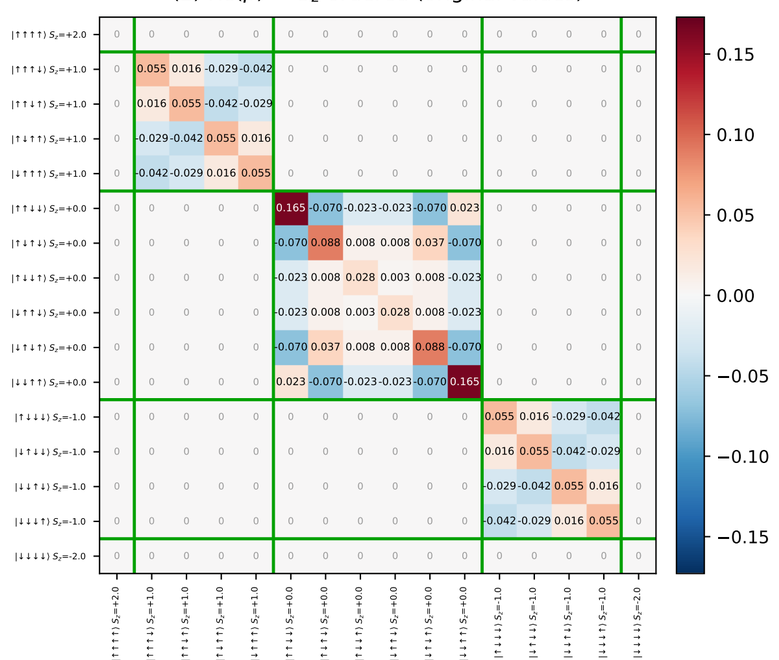


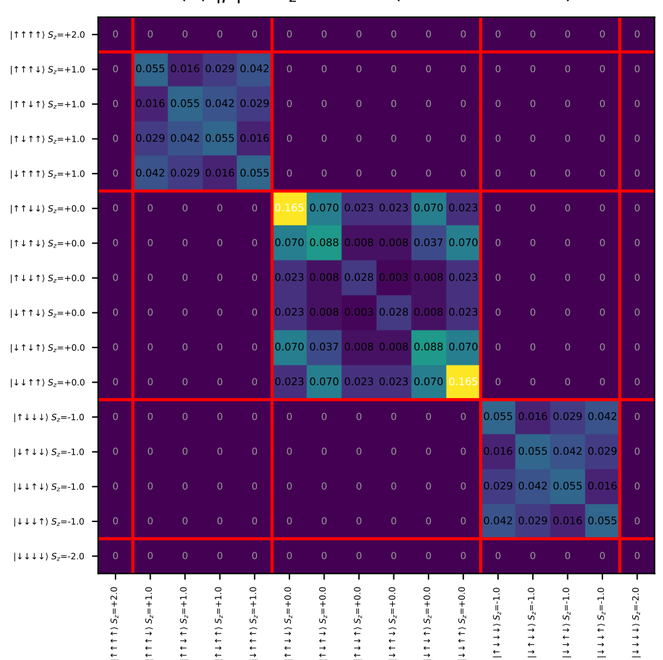
Trace=1.000000 Purity=0.178424 Entropy=1.920864 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.017324 ≤ C_{UB} = 0.432505 verdict: GME

4-Body RDM (Sites 0,2,3,5) — same matrix in all three views



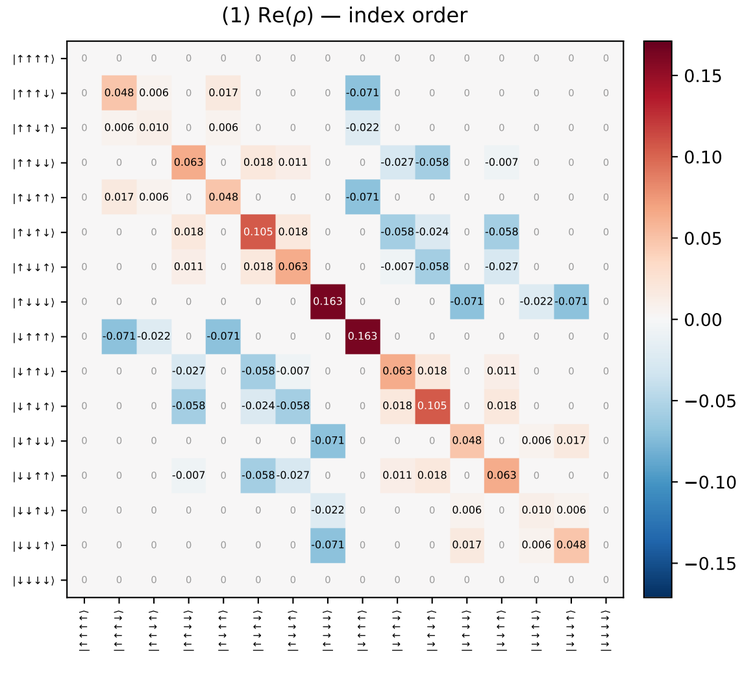


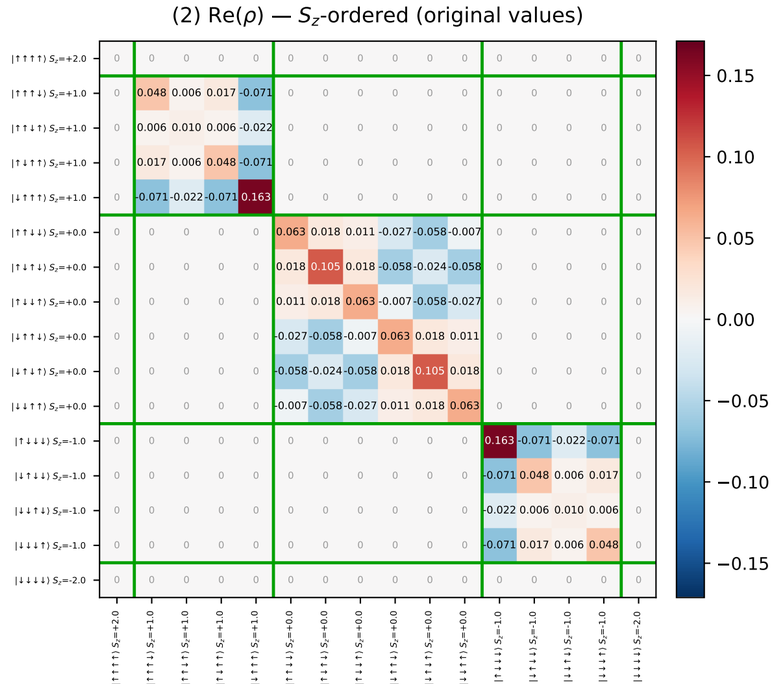


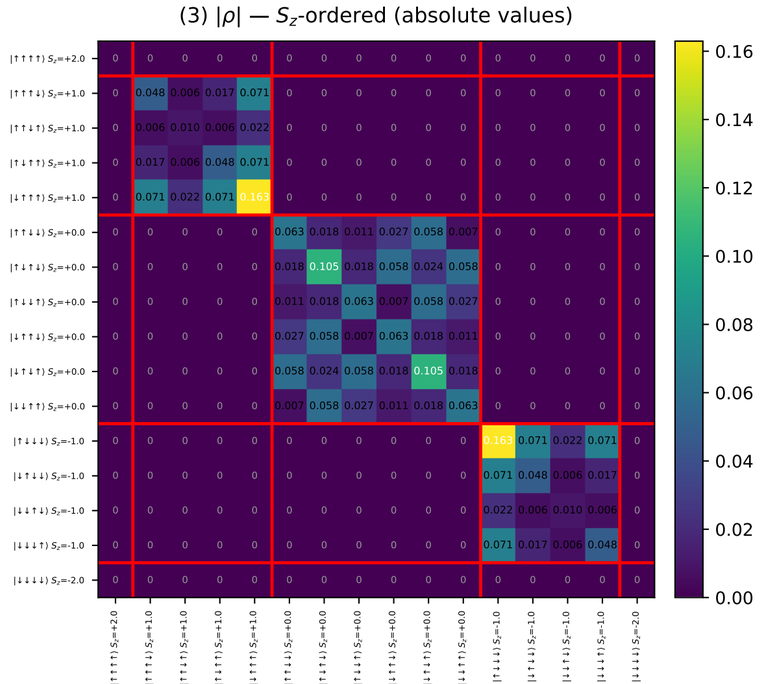
Trace=1.000000 Purity=0.166652 Entropy=2.049904 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.026478 ≤ C_{UB} = 0.740861 verdict: GME

4-Body RDM (Sites 0,2,4,5) — same matrix in all three views

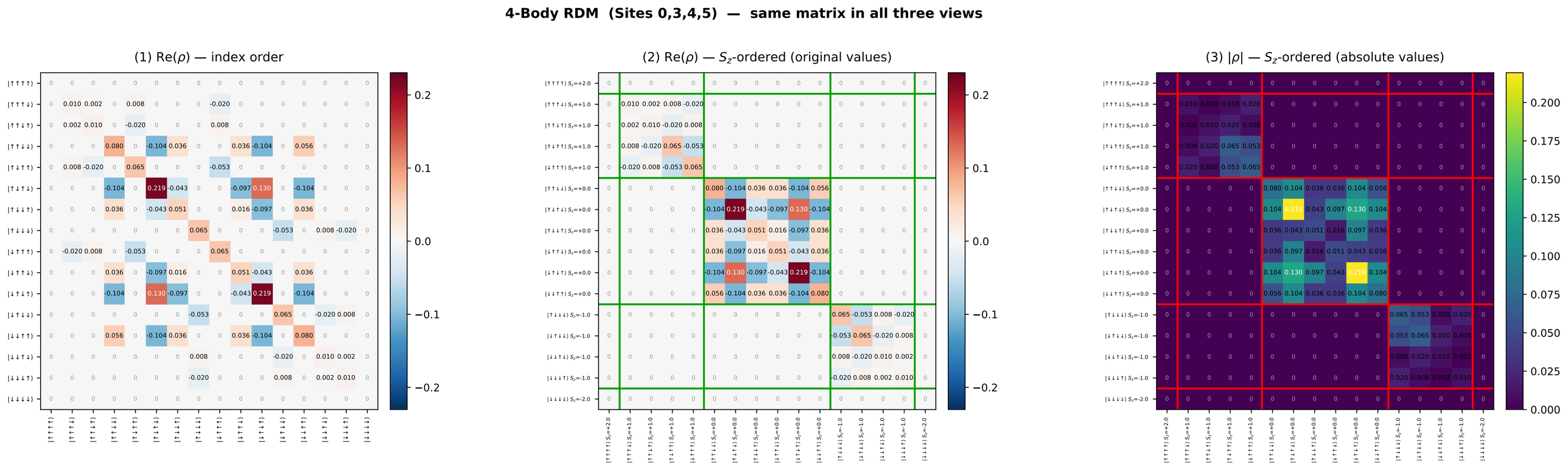




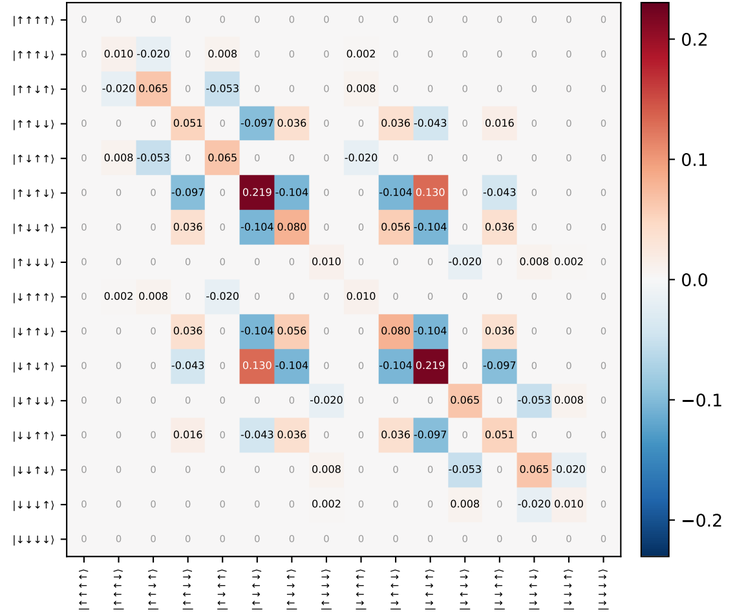


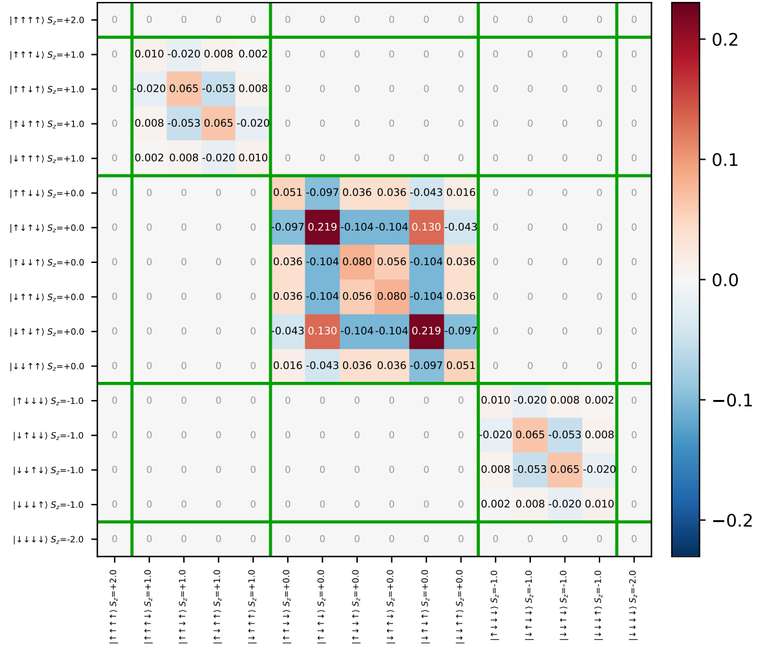
Trace=1.000000 Purity=0.178422 Entropy=1.920872 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

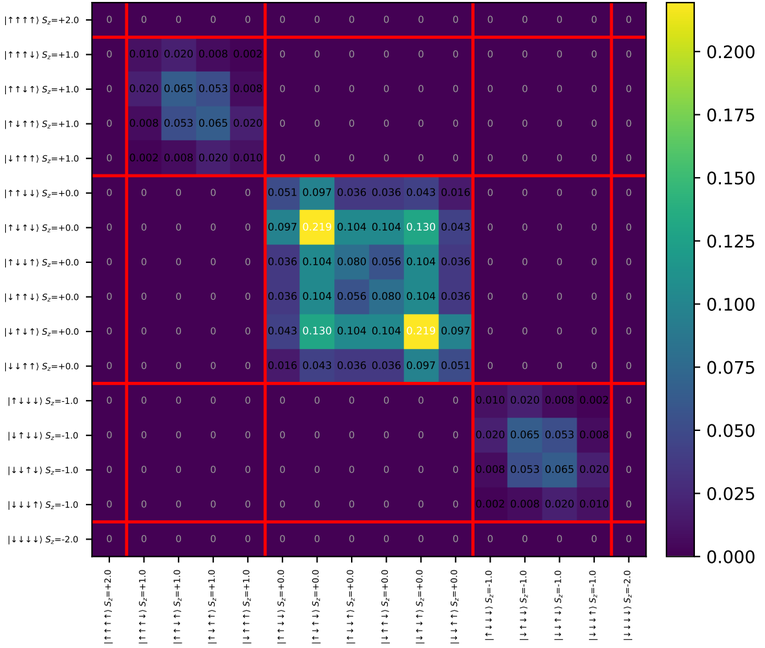
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.013390 ≤ C_{UB} = 0.432500 verdict: GME


4-Body RDM (Sites 0,3,4,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.329636 Entropy=1.492443 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.058692 ≤ C_{UB} = 0.603298 verdict: GME

4-Body RDM (Sites 1,2,3,4) — same matrix in all three views



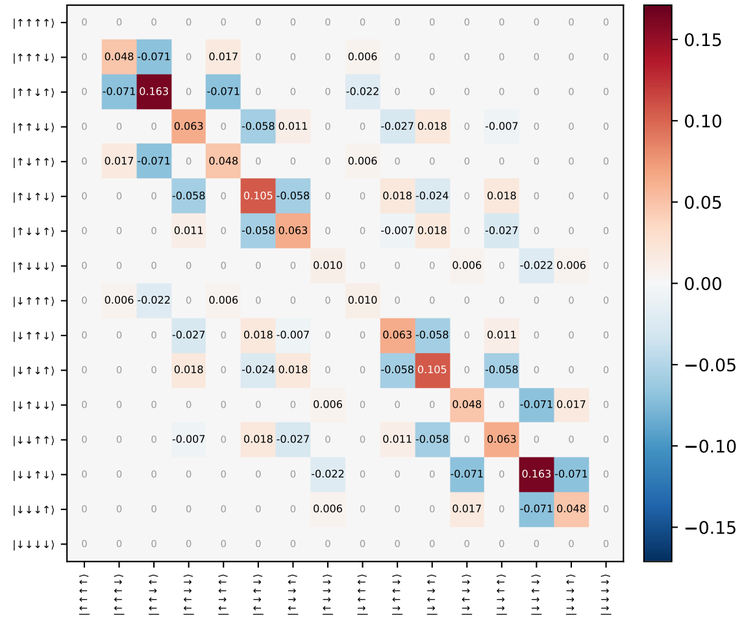


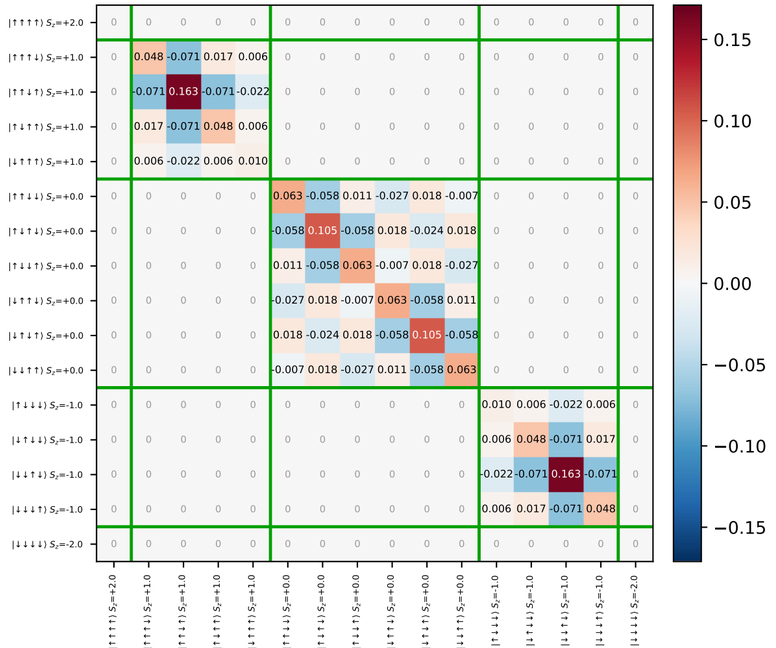


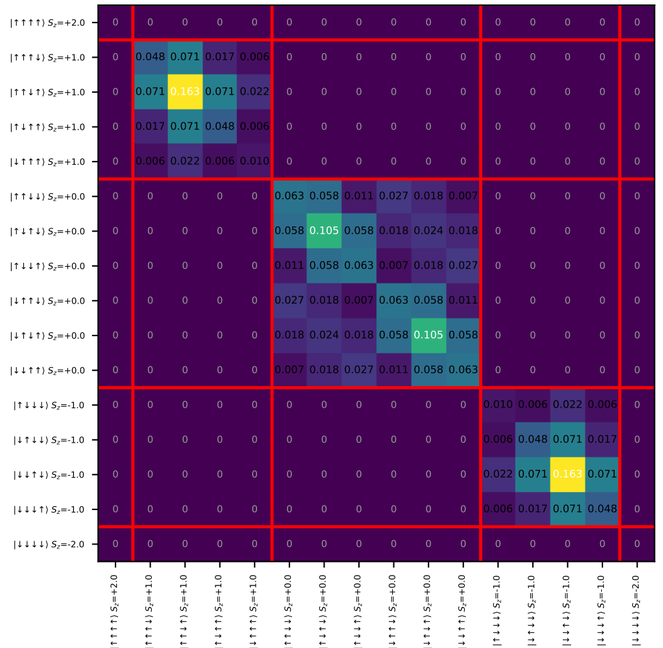
Trace=1.000000 Purity=0.329646 Entropy=1.492422 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.058444 ≤ C_{UB} = 0.603294 verdict: GME

4-Body RDM (Sites 1,2,3,5) — same matrix in all three views



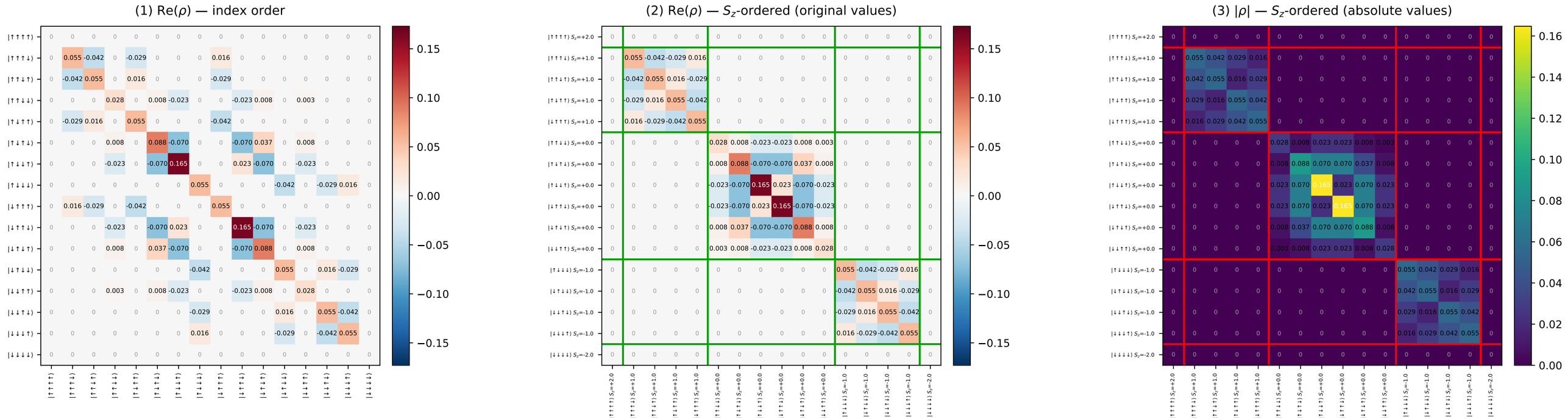




Trace=1.000000 Purity=0.178424 Entropy=1.920864 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.007188 ≤ C_{UB} = 0.432505 verdict: GME

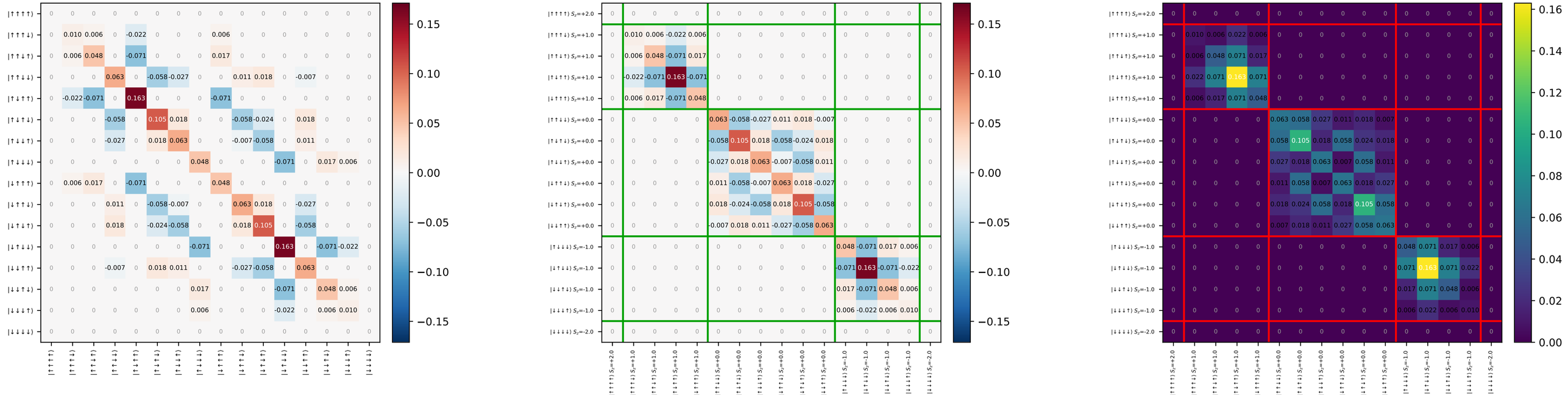
4-Body RDM (Sites 1,2,4,5) — same matrix in all three views



Trace=1.000000 Purity=0.166656 Entropy=2.049886 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.020211 ≤ C_{UB} = 0.740865 verdict: GME

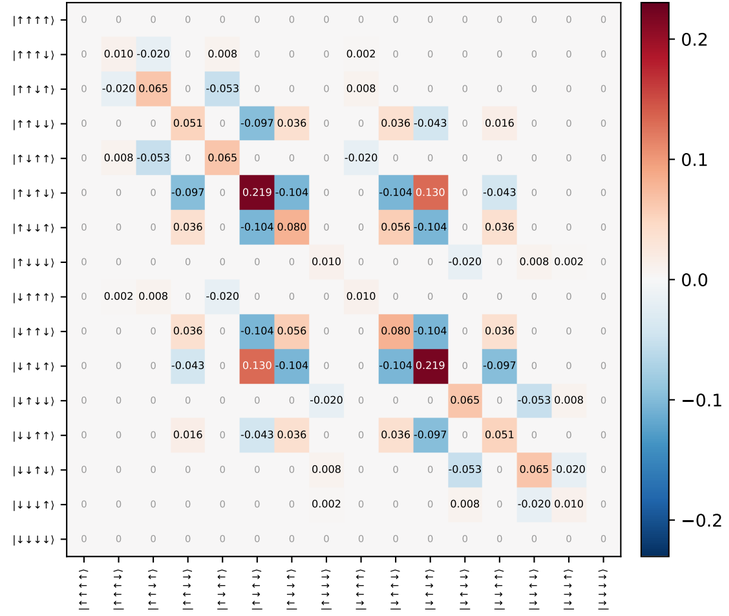
4-Body RDM (Sites 1,3,4,5) — same matrix in all three views

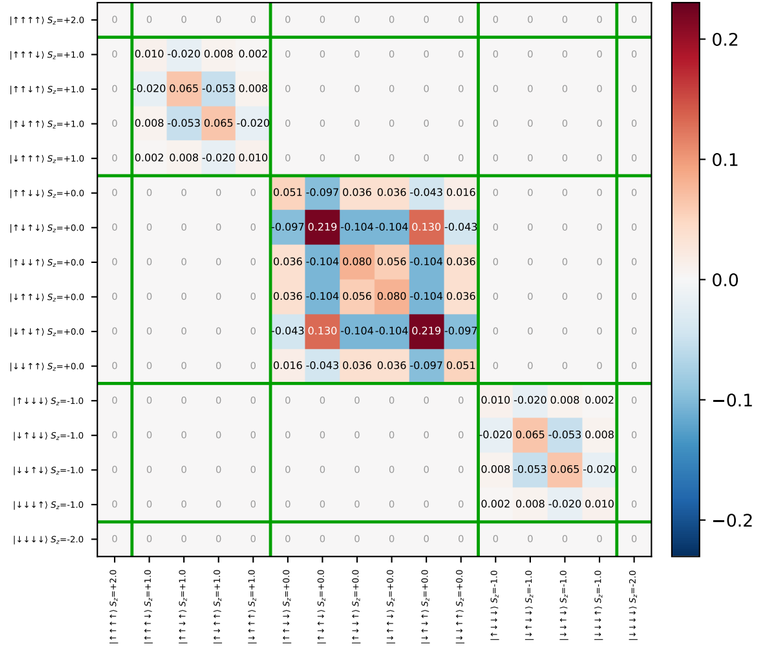


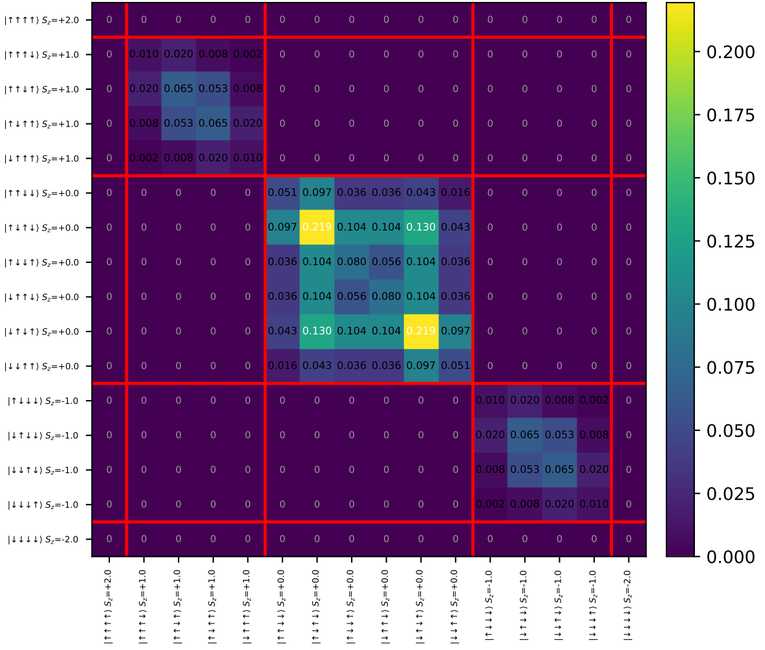
Trace=1.000000 Purity=0.178426 Entropy=1.920851 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} **= 0.000000** ≤ C_{GME} **= 0.013201** ≤ C_{UB} **= 0.432503** **verdict: GME**

4-Body RDM (Sites 2,3,4,5) — same matrix in all three views







Trace=1.000000 Purity=0.329644 Entropy=1.492429 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.058722 ≤ C_{UB} = 0.603289 verdict: GME